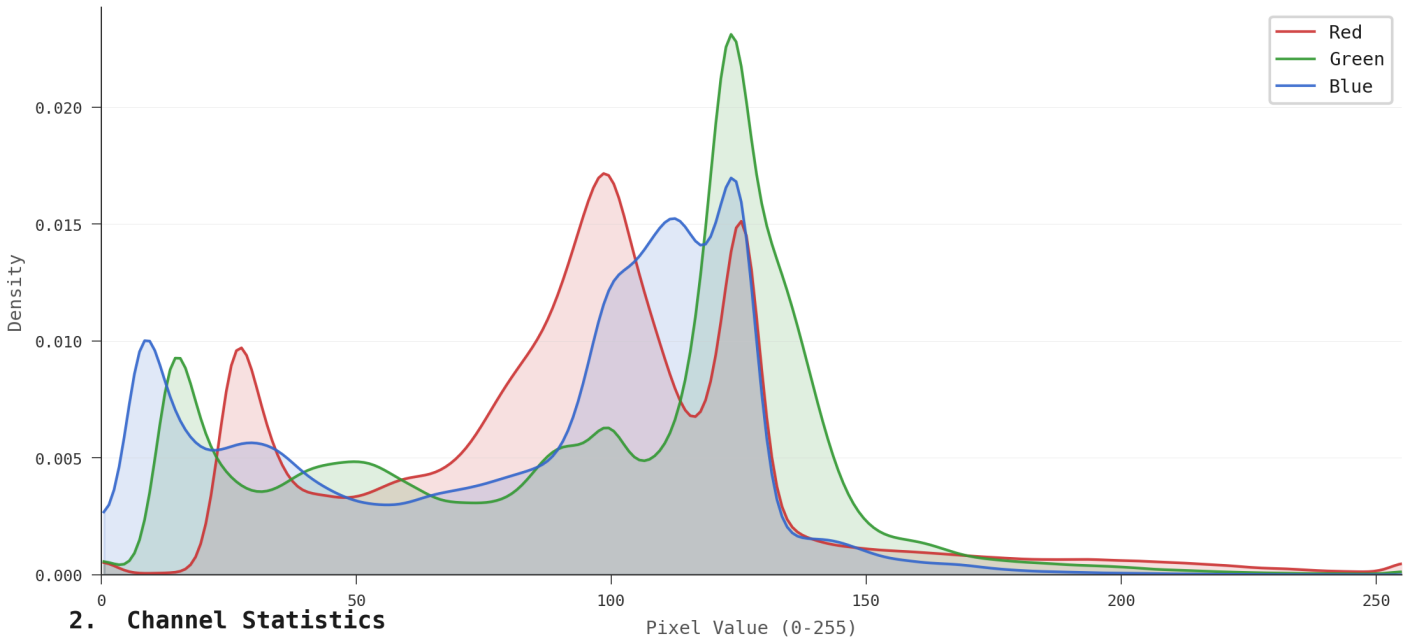
Reference Data Sheet: KODIM11

Subject: Zentime Pier and Dock

768x512 | 393,216 px
24-bit RGB | sRGB | PNG (lossless)

1. RGB Channel Distribution



2. Channel Statistics

Channel	Mean	Std Dev	Variance	Kurtosis	Skewness	Min	Max
Red	93.0274	39.8342	1586.7655	1.2927	0.4763	0	255
Green	95.0619	45.2329	2046.0129	-0.6104	-0.3633	0	255
Blue	82.3172	43.0273	1851.3476	-0.9188	-0.4345	0	255

3. Inter-Channel Correlation

	R	G	B
R	1.0000	0.8225	0.7906
G	0.8225	1.0000	0.9744
B	0.7906	0.9744	1.0000

Pairwise Pearson Correlation Coefficients

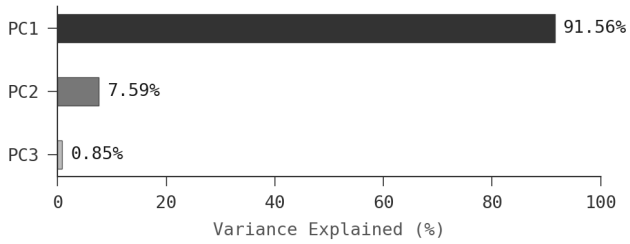
r(R-G)	=	**0.822536**
r(R-B)	=	**0.790610**
r(G-B)	=	**0.974415**
Avg |r|	=	**0.862520**

4. Covariance Matrix S

$$
\begin{array}{c} \\ R \\ G \\ B \end{array}
\begin{array}{c}
\begin{array}{ccc} R & G & B \end{array} \\
\begin{bmatrix}
\mathbf{1586.7695} & 1482.0621 & 1355.0760 \\
1482.0621 & \mathbf{2046.0181} & 1896.4575 \\
1355.0760 & 1896.4575 & \mathbf{1851.3523}
\end{bmatrix}
\end{array}
$$

Reference Data Sheet: KODIM11

5. Eigendecomposition



Eigenvector Loadings

	R	G	B	Eigenvalue
PC1	-0.5047	**-0.6286**	-0.5918	5021.4392
PC2	-0.8596	0.3027	0.4116	416.0334
PC3	0.0796	-0.7164	0.6931	46.6673

6. Derived Metrics

Condition Number

107.60

(l1/l3) PC1/PC2: 12.07 | PC2/PC3: 8.91

Blue Channel Independence

5.0%

Fraction of blue variance not captured by PC1

PC1 Dominant Channel

Green

Loading: -0.6286 | Green-Blue coupled

Classification: Weakly One-Dimensional (PC1 = 91.56%)

7. Spatial Autocorrelation (lag-1)

Channel	Horizontal	Vertical	Average
Red	0.936451	0.914191	0.925321
Green	0.950674	0.934059	0.942366
Blue	0.949338	0.932529	0.940933

8. Average Local Variance (3x3)

Channel	Value
Red	182.4999
Green	181.8437
Blue	163.2886

9. Redundancy Profile

Classification: Weakly One-Dimensional | Condition Number: 107.60 | PC1: 91.56%

Correlation: G–B pair dominates at r = 0.974415 | R–G: 0.822536 | R–B: 0.790610

Blue Channel: 5.0% independent | Moderate decoupling from PC1

Spatial: Avg autocorrelation: 0.936207 | Avg local variance: 175.8774

Eigenvector: Green-Blue coupled | G (-0.6286), B (-0.5918) co-load | R offset (-0.5047)

References

[1] Eastman Kodak Company. Kodak Publication No. PCD-042, 1992.

[2] Baetzel, J. (2026). "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."

[3] Watanabe, S. "Karhunen-Loeve Expansion and Factor Analysis," pp. 635-660, 1965.

[4] Giorgianni, E.J. and Madden, T.E. Digital Color Management. Addison-Wesley, 1998.

Blue Channel Independence = (1 - (blue_loading_PC1^2 x l1 / Var(B))) x 100

Data Source: Statistics derived from Baetzel (2026), "Statistical Characterization of Inter-Channel Redundancy Structure in the Kodak Lossless True Color Image Suite."